Electron-Ion Path Integral Monte Carlo with the Pair-Product Action

Riccardo Fantoni*
Università di Trieste, Dipartimento di Fisica, strada Costiera 11, 34151 Grignano (Trieste), Italy
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I. INTRODUCTION

[1]

In our reduced units we will measure lengths in units of a Bohr radius $a_B = \hbar^2/m_e e^2$ and energies in units of two Rydbergs e^2/a_B, i.e. of a Hartree. The Hamiltonian operator and the density matrix in reduced units will then read

$$H = K + V = -\sum_{\alpha=1}^{N} \lambda_{i_\alpha} \nabla_\alpha^2 + \sum_{\alpha<\beta=1}^{N} \phi_{i_\alpha i_\beta}(r_{\alpha\beta}), \tag{1.1}$$

$$\rho = e^{-H/T}. \tag{1.2}$$

where we indicate with a Greek index the particle label and with a Roman index the species label so that for example i_α is the species label of particle α, either 1 for the electrons or 2 for the protons, $\boldsymbol{r}_\alpha$ is the position vector of particle α, and $r_{\alpha\beta} = |\boldsymbol{r}_{\alpha\beta}| = |\boldsymbol{r}_\alpha - \boldsymbol{r}_\beta|$ is the distance between particles α and β. The partial pair potential is the usual Coulomb potential $\phi_{ij} = \epsilon_{ij}/r$ with $\epsilon_{ii} = +1$ for all i and $\epsilon_{ij} = -1$ for $i \neq j$. $\lambda_i = \hbar^2/2\mu_i$ where the reduced masses are $\mu_1 = 1$ and $\mu_2 = m_p/m_e \approx 1836.15$. Temperatures are in units of ≈ 315775K.

The coordinate representation of the high temperature density matrix is

$$\rho(R, R'; \tau) = \rho_0(R, R'; \tau) e^{-U(R,R';\tau)}, \tag{1.3}$$

$$\rho(R, R'; \tau) = \prod_{\alpha=1}^{N} (4\pi\lambda_{i_\alpha}\tau)^{-3/2} e^{-(\boldsymbol{r}_\alpha - \boldsymbol{r}'_\alpha)^2/4\lambda_{i_\alpha}\tau}, \tag{1.4}$$

where $\tau = \beta/M$ for large M. So that the coordinate representation of the thermal density matrix can be obtained from the following discretized path integral

$$\rho(R, R'; \beta) = \int \rho(R, R_1; \tau)\rho(R_1, R2; \tau) \cdots \rho(R_{M-1}, R'; \tau) \prod_{k=1}^{M-1} dR_k, \tag{1.5}$$

* riccardo.fantoni@scuola.istruzione.it

where $R_k = (\boldsymbol{r}_{1,k}, \boldsymbol{r}_{2,k}, \ldots, \boldsymbol{r}_{N,k})$ is the kth timeslice. Here $\boldsymbol{r}_{\alpha,k}$ is the position vector of particle α of species i_α at timeslice k in the path integral.

For the inter-action U we will adopt the end-point approximation for the pair-product action [1]

$$U(R, R'; \tau) \approx \sum_{\alpha<\beta} P_{i_\alpha i_\beta}(\boldsymbol{r}_{\alpha\beta}, \boldsymbol{r}_{\alpha\beta}; \tau), \tag{1.6}$$

where for P_{ij} we choose the pair Coulomb density matrix of Pollock [2]. Namely

$$P_{ij}(\boldsymbol{r}, \boldsymbol{r}'; \tau) = -\ln\left[\frac{\rho_{ij}(\boldsymbol{r}, \boldsymbol{r}'; \tau)}{\rho^0_{ij}(\boldsymbol{r}, \boldsymbol{r}'; \tau)}\right], \tag{1.7}$$

where

$$\frac{\partial \rho_{ij}(\boldsymbol{r}, \boldsymbol{r}'; \tau)}{\partial \tau} = \left[\lambda_{ij}\boldsymbol{\nabla} - \frac{\epsilon_{ij}}{r}\right]\rho_{ij}(\boldsymbol{r}, \boldsymbol{r}'; \tau). \tag{1.8}$$

Here $\lambda_{ij} = \lambda_i + \lambda_j$ and $\rho^0_{ij}(\boldsymbol{r}, \boldsymbol{r}'; \tau) = (4\pi\lambda_{ij}\tau)^{-3/2}\exp[-(\boldsymbol{r} - \boldsymbol{r}')^2/4\lambda_{ij}\tau]$ satisfies the same equation (1.8) but for $\epsilon_{ij} = 0$ for all i, j.

TABLE I. ...

case	statistics	T	M	n	N	x_2	μ_2	Ne_K	$-Ne_V$	$f_s(1)$	$f_s(2)$
A	b (PIMC)	0.07	20	0.3	30	0.5	1836.15	...	...	...	...
B	f (RPIMC)	0.07	20	0.3	20	0.5	1836.15	...	...	...	...
A	b (PIMC)	0.07	20	0.16	30	0.5	1836.15	...	...	...	...
B	f (RPIMC)	0.07	20	0.16	20	0.5	1836.15	...	...	...	...

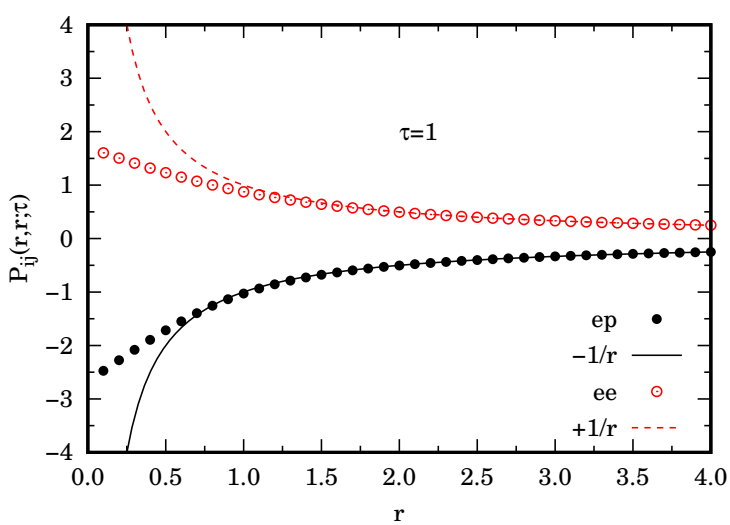

FIG. 1. We show the *ee* and *ep* pair Coulomb density matrix of Eq. (1.7) for $\tau = 1$. We compare it with the bare Coulomb potential. From the figure we can see that the cusp condition [2] $P_{ij}(\boldsymbol{r}, \boldsymbol{r}; \tau) \sim r\epsilon_{ij}/2\lambda_{ij}$ is satisfied at small r.

FIG. 2. C, D

Appendix A: Description of our PIMC and RPIMC algorithms

AUTHOR DECLARATIONS

Conflicts of interest

None declared.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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None declared.

[1] D. M. Ceperley, Path integrals in the theory of condensed Helium, Rev. Mod. Phys. **67**, 279 (1995).
[2] E. L. Pollock, Properties and computation of the Coulomb pair density matrix, Comput. Phys. Commun. **52**, 49 (1988).